A Constructive Introduction to Quantum Field Theory

Φ^4_3 on the torus II: a priori estimates and the limit

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SISSA lectures, Lecture 6 (provisional) — notes: Lecture 6, §§13–18

Where we are

At each lattice N, exactly (Lecture 5):

$$\mathrm{d}Y_q = [b_q(Y_q) + H_q(Y_q)\, Z_q]\mathrm{d}q + H_q(Y_q)\,\mathrm{d}W_q,$$
$$\mathrm{d}Z_q = -[b_q(Y_q) + H_q(Y_q)\, Z_q]\mathrm{d}q + \Xi_q\,\mathrm{d}W_q,$$

$$\psi = Y + Z = \int \mathrm{Cov}\,\mathrm{d}W.$$

Inputs.

A) Coercivity: $H_q(y) \asymp (K_q + \gamma y^2)^{-1}$, $\|H_q\|_{p\to p} \le (1+\varepsilon_q)/q$, $\varepsilon_q = C(g_q + g_q^2)$.

B) Source: $\|b_q^{\mathrm{b}}\|_\infty \le C\gamma^{7/2} q^{-5/2}$, $\|b_q^{\mathrm{I}}(y)\|_2^2 \le C\gamma^4 q^{-5} \|t_q(y)\|_6^6$.

C) Coordinates: $K_q(Y_q + \varrho_q(Y_q)) + \frac{\gamma}{3} Y_q^3 = P_q$, $\|D\,\varrho_q\|_{p\to p} \le \varepsilon_q$.

Goal

Z small and Y rarely at large fields, **uniformly in N**. Neither equation can be estimated alone.

How FBSDE estimates usually work, and why not here

Usual (Ma–Yong): Lipschitz coefficients, Itô on $\|Z\|^2$ and Gronwall; close on a short interval or under monotonicity.

Here: infinite interval, coefficients not uniformly Lipschitz, the coupling is not small. Instead we use **structure**:

- the backward equation is **linear in Z** given Y: a conditional representation with an explicit propagator;
- the forward equation is, by the field equation of Lecture 4, a cubic **elliptic** equation at each resolution: coercive, estimated **pathwise**;
- the two estimates are linked by two weighted energies with **matched weights**.

At fixed N, T all moments are finite (finite dimensions): used only to justify absorptions. Ma–Yong 1999

The energies

Lecture 6

The noise of the field equation

Lecture 5: $P_q = (q - q_0)\,\Phi_\infty + B_{q-q_0}$, a noisy measurement. Since $\psi_q = \mathbb{E}_q \Phi_\infty$,

$$\eta_q := P_q - (q - q_0)\,\psi_q = \mathbb{E}_q[B_{q-q_0}]:$$

the noise observable of Lecture 4 is the **filtered measurement noise**.

Lemma. *For convex Ψ, $\mathbb{E}\,\Psi(\eta_q) \le \mathbb{E}\,\Psi(B_{q-q_0})$, whatever the interaction. So $\zeta_q = C_q \eta_q$ has*

$$\mathbb{E}\,\mathrm{e}^{u\zeta_q(x)} \le \exp\left(\frac{u^2}{2}(q - q_0)\,a^2(q)\right), \qquad (q - q_0)\,a^2(q) \lesssim \sqrt{q}.$$

- Proof: conditional Jensen. One line.
- ζ is not Gaussian, but it has the **tails of the free field**, uniformly in N and in the interaction.
- Dudley + Doob: $\sup_q q^{-1/4}\,(\log(2+q))^{-1/2}\|\zeta_q\|_\infty$ has Gaussian tails.

The field equation

Insert $P_q = \eta_q + (q - q_0)\,(Y_q + Z_q)$ into the coordinate equation (C), with $K_q = K_{q_0} + q - q_0$. With $V_q = Y_q + \varrho_q(Y_q) - \zeta_q$:

$$K_{q_0} V_q + \frac{\gamma}{3} Y_q^3 = (q - q_0)\,(\zeta_q - \varrho_q(Y_q) + Z_q), \qquad Y_q = V_q + \zeta_q - \varrho_q(Y_q).$$

- The field equation of Lecture 4 in these coordinates: elliptic at each q, with **no memory**. The past enters only through the noise.
- The backward equation enters only through $(q - q_0)\,Z_q$, at the same resolution.
- The mass is only q_0: at large q all the coercivity must come from the **cube**.
- The identity is **pathwise**.

Exercise 13.3: subtracting the Gaussian field $X = \int C\,\mathrm{d}W$ instead gives an equation with memory.

Thresholds and energies

$$w_q = \operatorname{sgn}(V_q)\left(|V_q| - \tfrac{3}{4} r_q\right)_+, \qquad t_q = Y_q 1_{|Y_q| > r_q}, \qquad \xi_q = \zeta_q 1_{|\zeta_q| > r_q/32}.$$

$$E_T = \frac{1}{\gamma}\int q^{1/2} \|Z_q\|_2^2 \, \mathrm{d}q \qquad \text{backward energy}$$

$$J_T = \int q^{-3/2}\left[q\,\|w_q\|_4^4 + \gamma \|w_q\|_6^6 + \gamma \|t_q\|_6^6\right]\mathrm{d}q \qquad \text{large-field action}$$

$$G_T = \int \left[q^{-1/2}\|\xi_q\|_4^4 + \gamma\, q^{-3/2}\|\xi_q\|_6^6\right]\mathrm{d}q \qquad \text{noise tails}$$

quantity	weight	reason
$\|Z_q\|^2$	$q^{1/2}$	forcing: $q^{-3/2}\cdot\gamma^{-1}q^2\|Z\|_2^2$
$\gamma\|t_q\|_6^6$	$q^{-3/2}$	$q^{7/2}\|b^{\mathsf{I}}\|^2 \le C\gamma^3\cdot\gamma\, q^{-3/2}\|t\|_6^6$
$\|b_q^{\mathsf{I}}\|^2$	$q^{7/2}$	propagator s/q, $\int_q^\infty s^{-3/2} = 2q^{-1/2}$

Only **thresholded** sixth powers: unthresholded, $\mathbb{E}\|\zeta_q\|_6^6$ can be of order $q^{3/2}$ (free case), so $\int \gamma q^{-3/2}\,\mathbb{E}\|\zeta_q\|_6^6$ can reach γN^2, not uniform in N.

Forward coercivity

Lecture 6

The scalar model

Lemma. (scalar forward coercivity) *If* $a \geq 0$, $0 \leq p \leq q$ *and* $av + \frac{\gamma}{3}(v+g)^3 = p(g+z)$, *then, with* $r = (q/\gamma)^{1/2}$,

$$\frac{q}{8} w^4 + \frac{\gamma}{16} w^6 \leq 54 q \xi^4 + \frac{4225}{6} \gamma \xi^6 + \frac{12}{\gamma} q^2 z^2,$$

where $w = \mathrm{sgn}(v)\,(|v| - \frac{3}{4} r)_+$, $\xi = g\, 1_{|g| > r/32}$; *and* $|t| \leq \frac{32}{7}(|w| + |\xi|)$.

- The excess is controlled **at each resolution separately** by the noise tails and by the error Z.
- No smallness of the coupling, and no integration in q: multiply by $q^{-3/2}$ only to match the energies.

Proof: test with w^3

Where $w \neq 0$: $|v| = |w| + \frac{3}{4} r$, same sign, so $a\, v\, w^3 \geq 0$.

- **Cube.** $\operatorname{sgn}(w)\,(v+g) \geq |w| + \frac{11}{16} r - |\xi|$, and $\gamma r^2 = q$:

$$\frac{\gamma}{3} w^3 (v+g)^3 \geq \frac{\gamma}{12} w^6 + \frac{q}{4} w^4 + \frac{q\, r}{13} |w|^3 - \frac{4225}{6} \gamma\, \xi^6.$$

- **Noise.** $p\, g\, w^3 \leq \frac{q\, r}{32} |w|^3 + q|\xi||w|^3$: the first term is absorbed by the cube, and $q|\xi||w|^3 \leq \frac{q}{8} w^4 + 54\, q\, \xi^4$.
- **Error.** $q|z||w|^3 \leq \frac{\gamma}{48} w^6 + \frac{12}{\gamma} q^2 z^2$.

Above the threshold the cube provides a sixth power, a mass q and a cubic term which dominates the noise below the threshold.

Two elementary inequalities

For $x, \rho, u, r \geq 0$:

$$(x-\rho)^3 \geq \frac{3}{4}x^3 - 65\,\rho^3,$$

$$\frac{1}{3}u^3\left(u+\frac{11}{16}r-\rho\right)^3 \geq \frac{1}{12}u^6 + \frac{1}{4}r^2u^4 + \frac{1}{13}r^3u^3 - \frac{4225}{6}\rho^6.$$

Proof. $\frac{1}{4}x^3 + 64\,\rho^3 - 3\,x^2\rho = \frac{1}{4}(x-8\,\rho)^2(x+4\,\rho) \geq 0$. Then $x = u+b$, $b = \frac{11}{16}r$: $\frac{3}{4}b^2 > \frac{1}{4}r^2$, $\frac{1}{4}b^3 > \frac{1}{13}r^3$, and $\frac{65}{3}u^3\rho^3 \leq \frac{1}{6}u^6 + \frac{4225}{6}\rho^6$.

The lattice version

Proposition. (forward coercivity) *For q_0 large, at every N, pathwise and for every q,*

$$q\|w_q\|_4^4+\gamma\|w_q\|_6^6+\gamma\|t_q\|_6^6\le C\,(q\|\xi_q\|_4^4+\gamma\|\xi_q\|_6^6+\gamma^{-1}q^2\|Z_q\|_2^2).$$

So $J_T\le C\,(G_T+E_T)$, and $\|G_T\|_{L^p}\le C\gamma^3p^5$.

- **Elliptic term:** $w^3=\phi'(V)$, ϕ convex, so $\langle w^3,K_{q_0}V\rangle\ge 0$.
- **Coordinate correction:** $\varrho=\varrho^0+\varrho^1$, $\|\varrho^0\|_\infty\le\varepsilon r$ shifts the threshold, $\|\varrho^1\|_j\le C\,\varepsilon\,(\|w\|_j+\|\xi\|_j)$ is absorbed.
- **Noise tails:** $\mathbb{P}\,(|\zeta_q(x)|>r_q/32)\le 2\,\mathrm{e}^{-c\sqrt{q}/\gamma}$, by the lemma.

L^∞ coercivity

Proposition. *For q_0 large, at every N, pathwise and for every q,*

$$\|Y_q\|_\infty \le C\left[\left(\frac{q}{\gamma}\right)^{1/3}\|\zeta_q\|_\infty^{1/3} + \|\zeta_q\|_\infty + q^{1/4} + q\|Z_q\|_2\right].$$

So $t_q = 0$ whenever $\|\zeta_q\|_\infty + q\|Z_q\|_2 \le c\, r_q$.

- Remove the error: $\mathcal{H} = (q - q_0)\, C_{q_0} Z_q$, with $\|\mathcal{H}\|_\infty \lesssim q\|Z_q\|_2$.
- At a maximum of $W = V - \mathcal{H}$, the elliptic term has a sign: $\frac{\gamma}{3} Y^3 \le q\|d\|_\infty$. **The cube alone gives the bound.**
- **Large fields occur only when the noise or the error is large.**
- With $\|\zeta_q\|_\infty \lesssim q^{1/4}$ and $\|Z_q\|_2 \lesssim q^{-5/4}$ (Doob on $\mathbb{E}_q A^{\mathsf{I}}$): $\|Y_q\|_\infty \lesssim q^{5/12} \ll r_q$, up to logarithms.

Backward feedback and closure

Lecture 6

The backward representation

Propagator: $\partial_s \mathcal{P}_{q,s} = \mathcal{P}_{q,s} H_s(Y_s)$, $\mathcal{P}_{q,q} = I$, and by (A)

$$\|\mathcal{P}_{q,s}\| \le L_0 \frac{s}{q}, \qquad L_0 = \exp\left(\int_{q_0}^{\infty} \varepsilon_u \frac{\mathrm{d}u}{u}\right) \le \mathrm{e}^{C(g_{q_0} + g_{q_0}^2)}.$$

Lemma. (conditional representation)

$$Z_q = \mathbb{E}_q\left[\mathcal{P}_{q,T} Z_T + \int_q^T \mathcal{P}_{q,s}\, b_s(Y_s)\, \mathrm{d}s\right].$$

Proof: $\mathrm{d}_s(\mathcal{P}_{q,s} Z_s) = -\mathcal{P}_{q,s} b_s \mathrm{d}s + \mathcal{P}_{q,s} \Xi_s \mathrm{d}W_s$, a true martingale at fixed N.

Z = expected source on the unresolved scales, transported back, plus the boundary error. Compare $F_t = \mathbb{E}_t[F_T]$ in Lecture 4: here the source is the **defect**.

Three contributions

Bounded fields (two correction steps!):

$$\|Z_q^{\mathsf{b}}\| \le \int_q^\infty L_0 \frac{s}{q}\, C\gamma^{7/2} s^{-5/2} \mathrm{d}s = 2\, C L_0 \gamma^{7/2} q^{-3/2}, \qquad \frac{1}{\gamma}\int q^{1/2}\|Z^{\mathsf{b}}\|^2 \le C\gamma^6 q_0^{-3/2}.$$

Boundary: $q\|Z_q^{\mathsf{t}}\| \le L_0\, \mathbb{E}_q \|T Z_T\|$, contributing $\frac{2L_0^2}{\gamma q_0^{1/2}}\delta_{N,T,p}$, $\delta_{N,T,p} = \|T Z_T\|^2_{L^{2p}(\mathbb{P};L^2)}$.

Large fields: $A^{\mathsf{l}} = \int s^{7/2}\|b_s^{\mathsf{l}}\|^2 \le C\gamma^3 J_T$; Cauchy–Schwarz and Jensen,

$$\|Z_q^{\mathsf{l}}\|^2 \le 2L_0^2 q^{-5/2}\, \mathbb{E}_q A^{\mathsf{l}}, \qquad \left|\frac{1}{\gamma}\int q^{1/2}\|Z^{\mathsf{l}}\|^2\right|_{L^p} \le \frac{C\gamma^2}{q_0}\|J_T\|_{L^p}.$$

This is how the clipping error is controlled

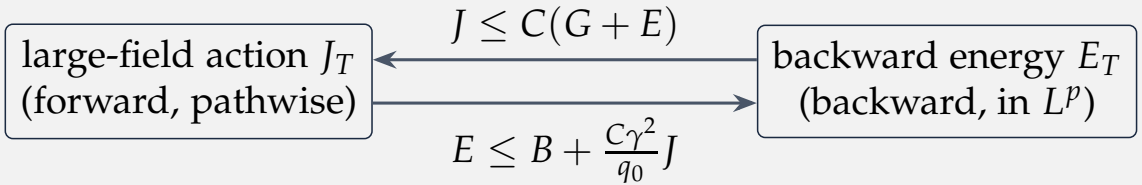
b^{l} is not small, but it enters the backward energy only through the large-field action J_T, with the small factor γ^2/q_0.

Closing the loop

Theorem. (cutoff-uniform energy estimate) *One $q_0 = q_0(\gamma)$: for all N, T, $p \geq 1$,*

$$\|E_T\|_{L^p} + \|J_T\|_{L^p} \leq C_\gamma\,(1 + p^5 + \delta_{N,T,p}).$$

$x = \|E_T\|_p$, $y = \|J_T\|_p$: $y \leq A_0\,(C\gamma^3 p^5 + x)$, $x \leq B_0 + B_1\,\delta + \eta\,y$, $\eta = C\gamma^2/q_0 = C g_{q_0}^2$. Choose $A_0\eta \leq \frac{1}{2}$; $y < \infty$ at fixed N, T, absorb.



The smallness is in the **feedback**, independent of p: one q_0 for all p; the growth p^5 is inherited from G_T.

Consequences

Once the boundary term is removed (Section 16):

$$\sup_N \|E_\infty^N\|_{L^p} + \sup_N \|J_\infty^N\|_{L^p} \le C_\gamma\,(1+p^5), \qquad \sup_N \mathbb{E}\|Z_q^N\|_2^2 \le C_\gamma\, q^{-5/2},$$

and $\mathbb{E}\exp\{c\,(E_\infty + J_\infty)^{1/5}\} < \infty$.

- The approximate forecast differs from the exact one by $O(q^{-5/4})$ in $L^2(\mathbb{P}; L^2)$ (root mean square), uniformly in the lattice.
- $H_q Z_q = O(q^{-9/4})$ in $L^2(\mathbb{P}; L^2)$: up to an integrable drift, the forward equation is the explicit SDE $\mathrm{d}Y = b(Y)\mathrm{d}q + H(Y)\mathrm{d}W$.

Comparison with the variational method

Comparison with Boué–Dupuis (Lecture 4)

There: drift $u = -4\lambda \dot{Q}^{1/2} {:}X^3{:} + r$ ($\lambda \int {:}\varphi^4{:}$ of Lecture 4). Here: $\bar{\Gamma}$ does all such changes of variables at once in the forecast coordinate; Z plays the role of r. Coercivity of the quartic: **pathwise** in the forward equation; the remainder: **in expectation** through the backward one.

Barashkov–G. 2020

Removing the cutoffs

So far, at every lattice N and upper scale T:

$$\|E_T\|_{L^p} + \|J_T\|_{L^p} \leq C_\gamma \left(1 + p^5 + \delta_{N,T,p}\right), \qquad \delta_{N,T,p} = \|T Z_T\|^2_{L^{2p}(\mathbb{P};L^2)}.$$

Three things remain:

1. the boundary term vanishes: $T Z_T \to 0$ at fixed N;
2. uniform energy bounds $\Rightarrow$ compactness of processes and measures;
3. identify the limit: coefficients, noise, equations.

The subtle point

Convergence of the fields does not pass an FBSDE to the limit: the backward martingale could acquire **extra noise**. The statistical meaning of the backward variable saves us.

The boundary at infinite resolution

Lecture 6

The boundary lemma

$\bar{\Gamma}_q$ was built to converge to the bare action: $\mu_q \to \mu_N$, $\Theta_q \to 0$, corrections integrated from $q = \infty$.

Lemma. (fixed N, constants depend on N)

$$\|Y_q - \Phi_\infty\|_{L^p} + \|q\, Z_q\|_{L^p} \le C_{N,p}\, q^{-1/2}, \qquad \textit{hence} \qquad \delta_{N,T,p} \le C_{N,p}\, T^{-1}.$$

- The field at resolution q: $\Phi_q = C_q P_q = (q - q_0)\, C_q\, \Phi_\infty + C_q B_{q-q_0} \to \Phi_\infty$.
- $\Phi_q = Y_q + C_q D\bar{\Gamma}_q(Y_q)$, so $Y_q - \Phi_q = O(q^{-1})$ and $D\bar{\Gamma}_q(Y_q) \to D\, V_N(\Phi_\infty)$.
- Exact minus approximate coordinate identity:

$$q\, Z_q = q\, C_q \left[D\bar{\Gamma}_q(Y_q) - \mathbb{E}_q D\, V_N(\Phi_\infty) \right];$$

compare $\mathbb{E}_q G$ with the **observed** $G_q^0 = D\, V_N(\Phi_q)$: $\|\mathbb{E}_q G - G\| \le 2\, \|G - G_q^0\|$.

No uniform estimate, no posterior concentration.

UV stability

Let $T \to \infty$ at fixed N (paths and ν_N do not depend on T; monotone convergence; $\delta_{N,T,p} \to 0$). **Only then** take $\sup_N$.

Theorem. (UV stability) *For $q_0 = q_0(\gamma)$ large,*

$$\sup_N \left(\|E_\infty^N\|_{L^p} + \|J_\infty^N\|_{L^p} \right) \le C_\gamma \left(1 + p^5\right), \qquad \sup_N \mathbb{E} \|Z_q^N\|_2^2 \le C_\gamma q^{-5/2},$$

and $\sup_N \mathbb{E} \exp\left\{c \left(E_\infty^N + J_\infty^N\right)^{1/5}\right\} < \infty$.

- $C_{N,p}$ diverges with N: never take $\sup_N$ of $C_{N,p}/T$.
- The force: $F_q = \mathbb{E}_q\left[-D V_N(\Phi_\infty)\right] = K_q Z_q - D\bar{\Gamma}_q(Y_q)$, the explicit force at the approximate forecast, corrected by $K_q Z_q$.

Compactness

Lecture 6

Suprema by a maximum principle

Embed by cells $\square_x$: $\iota_N : H_N \to H = L^2(\mathbb{T}^3)$ isometric, $j_N = \iota_N^*$.

Lemma. (supremum after closure) *Let* $\mathcal{Y} = \sup_q q^{-5/12} (\log (2+q))^{-1/6} \|Y_q\|_\infty$, $\mathcal{Z} = \sup_q q^{5/4} \|Z_q\|_2$. *Uniformly in* N,

$$\mathcal{Y} \le C_\gamma (1 + \mathcal{N} + \mathcal{Z}), \qquad \|\mathcal{N}\|_{L^p} \le C\sqrt{p}, \qquad \|\mathcal{Z}^2\|_{L^p} \le C_\gamma (1 + p^5).$$

- From the L^∞ coercivity above: **lattice maximum principle** in the field equation, no memory.
- $\mathcal{N}$: supremum of the noise (convex domination, Dudley, Doob). $\mathcal{Z}$: Doob on $\mathbb{E}_q A^{\mathrm{I}}$, after the boundary term is removed.
- The exponent $5/12 < \frac{1}{2}$ keeps Y below the threshold.

Finite-resolution suprema have all moments, uniformly in N. At the terminal resolution there is no such bound: Φ_∞ is a distribution.

Regularity and the terminal field

Space. $K_{q_0} V_q = f_q$ with $\mathbb{E}\int_I \|f\|_2^2 \le C_I$ (energies), so $V \in L^2(I;H^1)$; $Y = V + \zeta - \varrho(Y)$ has the regularity of the noise, that of the free field: H^σ, $\sigma < \frac{1}{2}$, in space; H^θ, $\theta < \frac{1}{2}$, in scale; tight in space–time L^p (and in $C(I;H)$: $C^{1/4}$ Hölder bound + uniform Fourier tails).

Terminal field. $H_q^2 \le \frac{C}{q} C_q$ gives, for $\frac{1}{2} < s < \frac{3}{2}$,

$$\int_Q^\infty \|(1-\Delta)^{-s/2} H_q\|_{\mathrm{HS}}^2 \mathrm{d}q \le C \int_Q^\infty \frac{1}{q} \sum_k \frac{(1+|k|^2)^{-s}}{q + c|k|^2} \mathrm{d}q \le C_s Q^{-(s-\frac{1}{2})},$$

and the drifts have squared tails $O(Q^{-3})$, $O(Q^{-5/2})$, $O(Q^{-3/2})$. Doob:

$$\sup_N \mathbb{E} \sup_{t \ge Q} \|Y_t - Y_Q\|_{H^{-s}}^2 \le C_s Q^{-(s-\frac{1}{2})}.$$

Corollary. *ν_N is tight in $H^{-s}(\mathbb{T}^3)$ for every $s > \frac{1}{2}$, with all moments uniformly in N. (Threshold of the free field in $d = 3$.)*

Identifying the limit

Lecture 6

The deterministic coefficients

Uniform bounds do not identify a limit: a weakly convergent vertex cannot in general be contracted against a varying covariance. Three tools:

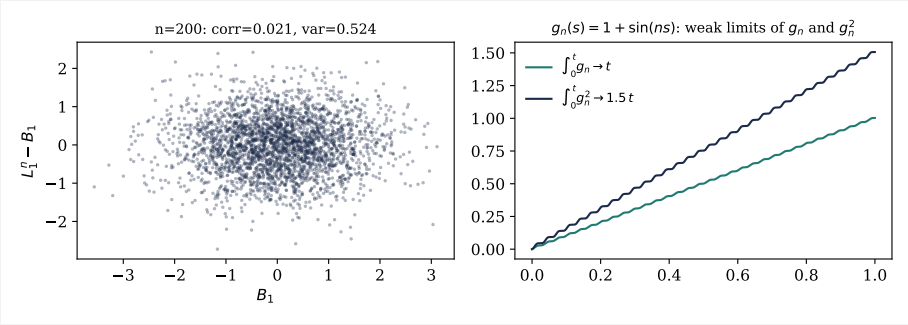
- **Weak vertex measures**: bounded rooted norms $\Rightarrow$ signed measures, weakly compact; the centred kernel in a first-moment topology. The limits solve the same fixed point, unique in its ball: the whole sequence converges.
- **Compact sandwich**: $G_N \to G$ in HS, $\mathsf{B}_N \to \mathsf{B}$ weakly, symmetric, uniformly bounded, $\sup_N \|G_N \mathsf{B}_N\| \le \theta < 1$, $\|G\mathsf{B}\| \le \theta \Rightarrow (I + G_N \mathsf{B}_N)^{-1} G_N \to (I + G\mathsf{B})^{-1} G$ in HS.
- **Contraction against a moving covariance**: rows of the inverses are Hölder + HS convergence $\Rightarrow (H^N)^2 \to H^2$ uniformly; weak vertices can be contracted.

Hence $H_q^N(Y_q^N) \to H_q(Y_q)$ in $L^2(I;\mathsf{HS})$ and $b_q^N(Y_q^N) \to b_q(Y_q)$ in $L^2(I;H)$.

Hidden noise

$L_t^n = \int_0^t g_n(s)\,\mathrm{d}B_s$, $g_n = 1 + \sin(ns)$: integrands $\to 1$ weakly, squares $\to \frac{3}{2}$.

$$(L^n, B) \Longrightarrow (B + 2^{-1/2}\tilde{B},\ B), \qquad \tilde{B} \perp B.$$



Left: $(B_1, L_1^n - B_1)$, $n = 200$: correlation ≈ 0, variance of the gap $\approx \frac{1}{2}$ — the residual does not vanish and is orthogonal to B. Right: $\int_0^t g_n \to t$ but $\int_0^t g_n^2 \to \frac{3}{2}t$.

A limiting FBSDE obtained this way would carry an **extra martingale** orthogonal to the driving noise. We need **strong** compactness of the backward integrand $\Xi = \mathrm{Cov}\text{-}H(Y)$.

The Riccati equation of the posterior covariance

Lemma. $\mathrm{Cov}_q = D^2\,\mathscr{K}_q(P_q) \geq 0$ *and*

$$\mathrm{dCov}_q = -\mathrm{Cov}_q^2\,\mathrm{d}q + D^3\,\mathscr{K}_q(P_q)[\mathrm{d}W_q]$$

Lecture 4: conditional variance is resolved at rate $\mathrm{Cov}_t\dot{\mathscr{R}}_t\mathrm{Cov}_t$; here $\partial_q\mathscr{R} = 1$.

- $\langle f, \mathrm{Cov}_q f\rangle \geq 0$: a **nonnegative supermartingale**, initial value $\mathrm{Var}_{\nu_N}\langle \Phi_\infty, f\rangle$ bounded uniformly (uniform moments, Corollary).
- Stop at level K: decreasing part and bracket have bounded expectations $\Rightarrow$ time regularity $W^{a,1}$, $a < \frac{1}{2}$ $\Rightarrow$ strong compactness of each finite block.
- Spatial tails: isometry + symmetry, and terminal averaging $\psi_Q = \mathbb{E}_Q \int_L^{L+1} (Y_s + Z_s)\mathrm{d}s$ (H^σ bound on Y, $\mathbb{E}\|Z_s\|^2 \leq C s^{-5/2}$).

$\Rightarrow$ Cov^N tight in strong $L^2(I; \mathrm{HS})$; brackets $\int \mathrm{CovCov}^*$ and covariations with W converge.

Identification of the martingales

On a common filtered space (Skorokhod), along a subsequence:

- ψ is a continuous square-integrable martingale for the joint filtration; its bracket and its covariation with W are those of $\int \mathrm{Cov}\,\mathrm{d}W$;
- so $\psi - \int \mathrm{Cov}\,\mathrm{d}W$ has zero bracket and vanishes: **no extra martingale**;
- same for the forward equation, and $X = \int C\,\mathrm{d}W$ with the **same** W.

$$Y_q = \int_{q_0}^{q} [b(Y) + H(Y)\,Z]\mathrm{d}s + \int_{q_0}^{q} H(Y)\,\mathrm{d}W,$$

$$Z_q = -\int_{q_0}^{q} [b(Y) + H(Y)\,Z]\mathrm{d}s + \int_{q_0}^{q} \Xi\,\mathrm{d}W, \qquad X_q = \int_{q_0}^{q} C_s\,\mathrm{d}W_s.$$

The result (I): convergence and the limiting FBSDE

Theorem. (subsequential construction) *For $\lambda > 0$, q_0 large, along a subsequence of lattices:*

a) (Y^N, Z^N, ψ^N, X^N), *with* $X^N = \int C \, \mathrm{d}W^N$, *converge in* $C(I;H)$, W^N *in* $C(I; H^{-r})$, $r > \frac{3}{2}$, (Cov^N, Ξ^N) *in strong* $L^2(I;\mathrm{HS})$, *jointly, a.s. on a new space;*

b) the limit solves the FBSDE on $[q_0, \infty)$ *with the continuum coefficients* b, H *(unique in their ball), an* H*-cylindrical Brownian motion obtained as the limit of the lattice innovations,* $\mathrm{Cov} = H(Y) + \Xi \geq 0$, *and* $\mathbb{E}\|Z_q\|^2 \leq C q^{-5/2}$;

ψ^N is tight in $C(I;H)$ because its brackets $\int \mathrm{Cov}^N \mathrm{Cov}^{N*}$ are tight and its Fourier tails are uniformly small.

The result (II): the terminal law

Theorem. (subsequential construction, continued)

a) $Y_q \to \Phi$ *in* H^{-s}, $s > \frac{1}{2}$, *a.s. and in mean square;* $\mathrm{Law}(\Phi) = \nu$ *is the weak limit of* ν_N, *and* $\langle Y_q + Z_q, h\rangle = \mathbb{E}[\langle \Phi, h\rangle \mid \mathcal{F}_q]$;

b) the limiting thresholded energies (built with $\zeta_q = C_q(W_q + \int \psi - (q - q_0)\psi_q)$*) obey the bounds of UV stability.*

The posterior structure survives: $\psi = Y + Z$ is a **closed observable**, Y is computed from the explicit action, Z is small at high resolution.

The renormalised field equation

Theorem. (integration by parts) $G(\varphi) = g(\langle \varphi, h_1 \rangle, \dots, \langle \varphi, h_k \rangle)$, $g \in C_b^\infty$, h, h_i *smooth:*

$$\lim_N \mathbb{E}\left[G\left(\iota_N \Phi_\infty^N\right) \langle \mu_N \Phi_\infty^N + \lambda (\Phi_\infty^N)^3, j_N h \rangle \right]$$

$$= \mathbb{E}_\nu \left[D_h G(\Phi) - G(\Phi) \langle \Phi, (q_0 - \Delta) h \rangle \right].$$

Proof: Gaussian integration by parts in $\mathrm{e}^{-V_N - \frac{1}{2}\langle \varphi, K_{q_0} \varphi \rangle}$; tightness and uniform moments.

- The terminal case of the DS equations of Lecture 4.
- The renormalised cube $\mu_N \Phi + \lambda \Phi^3$ is identified only **tested in expectation**, not as a random distribution.
- Limiting force without terminal representative: **not proved**.
- At every finite resolution the force is explicit: $F_q = K_q Z_q - D\bar{\Gamma}_q(Y_q)$.

The field equation at resolution q

In the limit: $\Phi_q = C_q P_q$, $\eta_q = P_q - (q - q_0)\,\psi_q$, $F_q = K_q \psi_q - P_q$, $J_q = (q - q_0)\,C_q \to I$.

Theorem. *For every q, with no renormalisation constant, pathwise as distributions,*

$$K_{q_0}\Phi_q = J_q F_q + \eta_q, \qquad \mathbb{E}\, D_{J_q h} G(\Phi_q) = \mathbb{E}\big[G(\Phi_q)\langle \eta_q, h\rangle\big].$$

$\langle F, h\rangle$, $\langle \eta, h\rangle$ are martingales, and $\|\langle F_q + D\bar{\Gamma}_q(Y_q), h\rangle\|_{L^2} \lesssim \|h\|_{H^2} q^{-1/4}$.

- **Proof of DS:** $\eta_q = \mathbb{E}_q B_{q-q_0}$ and $\Phi_q = J_q \Phi_\infty + C_q B_{q-q_0}$: Gaussian integration by parts in B, then $N \to \infty$.
- **The renormalised cube is an observable:** the unique martingale in the class of the explicit $-D\bar{\Gamma}_q(Y_q)$.
- $q \to \infty$ recovers the integration by parts formula. Not proved: that F is closed.

Exponential integrability: the tilt

Hairer–Steele. $\hat{\nu} = \mathcal{Z}^{-1} e^{W} \nu$. For every set A:

$$\mathcal{Z} \le \exp(\sup_{A} W) \, / \, \hat{\nu}(A).$$

It is enough that the tilted measure gives mass $\frac{1}{2}$ to a fixed set. They use the dynamics: for $W = \frac{\beta}{4}\langle \varphi, \psi \rangle^4$ the force $\beta \psi \langle \Phi, \psi \rangle^3$ is below the cube, and coming down from infinity is uniform.

In the filter. Same channel, same $\bar{\Gamma}$, prior $\hat{\nu}_N$:

- the same b, H, the same noise bound, the same forward and backward estimates;
- only the boundary changes: $T\hat{Z}_T \to D\,W(\hat{\Phi})$, so

$$\|\hat{E}_\infty\|_{L^p} + \|\hat{J}_\infty\|_{L^p} \le C_\gamma (1 + p^5 + \|D\,W(\hat{\Phi})\|^2_{L^{2p}(L^2)}).$$

Closing: the scale adapts to the level

Test the coordinate equation with ψ, $X = \langle \hat{\Phi}, \psi \rangle$, $P_q = (q - q_0)\hat{\Phi} + B$:

$$|X| \le C(r_q + \|t_q\|_6 + \frac{\gamma}{q}\|t_q\|_6^3 + \mathcal{N}_\psi).$$

Choose Q with $r_Q \sim |X|$, so $Q \sim \gamma X^2$: then $\|t_q\|_6 \gtrsim |X|$ on $[Q/2, Q]$, and

$$|X|^5 \lesssim 1 + \mathcal{N}_\psi^5 + \hat{J}_\infty.$$

So $W = \beta(1 + X^2)^{7/4}$ closes for small β, uniformly in N:

$$\mathbb{E}\exp(\beta|\langle \Phi, \psi \rangle|^{7/2}) \le C, \quad \mathbb{E}\exp(\beta\|\Phi\|_{H^{-s}}^2) \le C \quad (s > \frac{1}{2}).$$

Not quartic: on one site $\hat{J} \sim |X|^5$ is sharp, while $\|DW\|^2 \sim \beta^2|X|^6$. The backward bound s/q ignores that $H(Y)$ is small on large fields.

Scale-by-scale DS and uniqueness

DS along the scales. For $G = g(\langle \Phi_{q_1}, h_1 \rangle, \dots, \langle \Phi_{q_k}, h_k \rangle)$, $q_i \le q$:

$$\mathbb{E} \sum_i \partial_i g \langle J_{q_i} h, h_i \rangle = \mathbb{E}[G \langle \eta_q, h \rangle] = \mathbb{E}[G(\langle \Phi_q, K_{q_0} h \rangle + \langle D\bar{\Gamma}_q(Y_q), J_q h \rangle)] + O(q^{-1/4}).$$

Uniqueness in finite volume, conditional (as in Hairer–Steele, Thm 2.2). ν_N is invariant for the lattice equation with $\mu_N = m_N + c_N$. Assume lattice convergence, ergodicity and tightness in our setting: if $c_N \to c$, then $\nu_N \to \pi_c$, the unique invariant measure.

- The literature covers dyadic grids, unit coefficients and a fixed finite part; our odd N and varying c_N are assumed extensions.
- If $\nu = \pi_c$, quartic tails hold: $\mathbb{E}_\nu \exp(\beta \langle \Phi, \psi \rangle^4) < \infty$ (Hairer–Steele).
- **Not proved here:** convergence of c_N (lattice sums); uniqueness for the FBSDE as an equation; an intrinsic proof.

What is not proved

- **Uniqueness** for the FBSDE as an equation; convergence of the finite part of the counterterm; an intrinsic proof.
- **Small mass / strong coupling**: the construction needs $\gamma\, q_0^{-1/2}$ small, i.e. weak coupling $\lambda / m \ll 1$. Removing it needs a priori estimates without the small feedback γ^2 / q_0, and a coefficient construction outside the perturbative ball.
- **Axioms**: reflection positivity holds on the lattice (lattice and mid-lattice planes) and passes to ν for every coordinate reflection (standard; exercise); rotation invariance and clustering need infinite volume.
- **Infinite volume**: the force and its Burgers flow are intensive; estimates must be made local. Uniform-in-volume coefficient bounds and local energy moments are open; retaining the full FBSDE also needs control of the **noise from infinity** in the nonlocal Ξ.

Summary of the course

Wightman → Euclidean (L1) → free field (L2) → stochastic quantisation, scale decomposition (L3) → Polchinski, FBSDE, variational problem (L4) → Φ^4_3 through the FBSDE in forecast coordinates (L5–6).

Lecture 4 (formal)	Lectures 5–6	in the limit
forecast $\psi_t = \mathbb{E}_t \Phi_T$	$\psi = Y + Z$	closed martingale
response $H_t = \mathrm{Cov}_t$	$\mathrm{Cov} = H(Y) + \Xi$	strong $L^2(\mathrm{HS})$, Riccati
Γ_t, Wetterich	$\bar{\Gamma}_q$ explicit, defect e	deterministic coefficients
force F_t	$K_q Z_q - D\bar{\Gamma}_q(Y_q)$	observable at finite q
DS equation	integration by parts	tested in expectation

UV stability = an explicit action predicts the conditional mean up to an error that is small at high resolution.